Learner Workbook

Applied AI and Automation for Executives. One page for each session. Write in it, or type in Word.

How to use this workbook

Each page has the day's question, your steps for the paper exercise or practical, the fields to fill in, two questions to think about and one decision.
Sessions 1 to 11 each have a worksheet page after the session page, with the numbers for the paper exercise. The templates named on a page are in your Templates folder and the case company files are in Practice Files. The answers to class exercises are not in this workbook.
From Session 12 you work on your own workflow and organisation. Before then, use the case company for the exercises.

My workflow (fill in at Session 12)

The workflow or email task I will map	
Owner	
How often and how long it takes today	
What better means (a number)	
AI tool I already use	

Session 0, the introduction

AI and machine learning: the map before the route

What do I need to understand about AI before I decide anything about it?

Your steps for today

- Read the Course Handbook before Session 1.
- Write down which part of the 14-session map you most want to understand.
- Write any question from today that was not answered.

Fill in

The session I most want to understand	
One idea from today I could explain to a colleague	
A question from today that I still have	
My laptop and AI assistant account are ready (yes or no)	

Think about

Which field or industry use on today's slides is closest to my company?

Which role in the line from data to decision is missing in my company?

My decision

The one question I want answered first in the course.	

I keep: The Course Handbook, with today's ideas, a glossary and what to prepare. Read it before Session 1, which begins with how a machine sees.

Session 1

Seeing: pixels, features and neurons

How can a machine read a picture when all it has is numbers?

Your steps for today

- Price the five houses on the worksheet with weights 2 and 10 and bias 30.
- Change the weight of a bedroom to 15 and write how much each price moves.
- Compute the weighted sum, add the bias −6 and apply the step for each of the five days.
- Change the bias to −7 and mark which day changes.
- Write one place in your work where pictures or scans are read, and one check you would keep.

Fill in

Price of each of the five houses	
New price of each house with a bedroom weight of 15	
Weighted sum, total with bias and decision for each of the five days	
Which day changed when the bias became -7	
One place in my work where pictures or scans are read	
The check I would keep	

Think about

What would I ask a vendor who says their system reads our scans accurately?

Which of our images are unusual or poor, and would a system have seen those?

My decision

Use it now, use it with a checker, wait, or do not use it, with the reason.	

I keep: A one-page sheet: how a picture becomes numbers, a neuron worked by hand on a house price and an umbrella, and the digit results.

Session 1: worksheet for the paper exercise

A neuron prices houses and decides on an umbrella

Part 1: a neuron prices a house. Weighted sum = floor area x 2 + bedrooms x 10. Add the bias of 30 to get the price in UGX million.

House	Floor area (m²)	Beds	Area x 2	Bedrooms x 10	Bias	Price (UGX m)	Price if a bedroom is 15
A	80	3			30		
B	50	2			30		
C	120	4			30		
D	65	2			30		
E	100	3			30		

Now make each bedroom worth 15 instead of 10. By how much does every price move? _____

Part 2: a neuron decides on an umbrella. Weighted sum = 0.1 x chance of rain (%) + 0.05 x minutes outside. Total = weighted sum + bias. Activation function (step): a total above 0 gives 1, carry an umbrella.

Day	Rain %	Minutes	0.1 x rain	0.05 x minutes	Weighted sum	Total, bias -6	Carry? Y/N	Total, bias -7	Carry? Y/N
Mon	70	30							
Tue	20	30							
Wed	50	60							
Thu	40	20							
Fri	60	10							

Which day changed when the bias went from −6 to −7? _____ What do a weight and a bias do, in one sentence?

_____

_____

Session 2

Learning from examples: types of machine learning, decision trees, nearest neighbours

How does a machine learn a rule from examples, and how much can I trust what it learned?

Your steps for today

- Fill in the distance table for the 10 sold houses and the new house.
- Average the prices of the 3 nearest (k = 3), then use only the nearest (k = 1), and write both estimates.
- Say which estimate you would quote to a buyer, and why.
- Write one recurring decision in your work and how many past cases you have on record.

Fill in

Distance from the new house to each sold house	
The estimate with k = 3, and with k = 1	
The estimate I would quote, and why	
A recurring decision in my work	
How many past cases I have on record	

Think about

Which would my auditor accept more easily: a rule or similar past cases?

Do I have enough past cases, and are they labelled?

My decision

A tree, neighbours, neither or a person, with the number of past cases.	

I keep: A one-page sheet: how a tree picks a question, overfitting and cross-validation, distance and k-nearest neighbours, and the honest scores from today's run.

Session 2: worksheet for the paper exercise

Estimate a house price from its nearest neighbours

Estimate the price of the new house N from the sold houses most like it. One square across is 20 m² of floor area, so N (5 across, 4 up) is 100 m². One square up is 1 km to the town centre, so N is 4 km out. Distance is the Manhattan count: squares across plus squares up.

House	Across	Up	Distance (across + up from N)	Price (UGX million)
A	2	1		80
B	3	2		95
C	4	1		120
D	4	5		100
E	5	6		115
F	7	3		175
G	6	4		150
H	8	6		180
I	7	2		180
J	9	3		220

k = 3: the three nearest houses		**Average of their prices**	
k = 1: the nearest house		**Price estimate**	
Gap between the two estimates		**Which would I quote to a buyer?**	

Draw the houses on squared paper (across along the bottom, up along the side), write each price beside its point, and circle the 3 nearest to N.

Session 3

Measuring a model: accuracy, precision, recall, F1, ROC and AUC, thresholds, baselines, overfitting

How do I know whether a model is any good, and what should I ask the vendor who says it is?

Your steps for today

- The smoke alarm confusion matrix (TP 16, FP 34, FN 4, TN 946): work out accuracy, precision, recall and F1 in the margin.
- The vendor question sheet: read all ten questions.
- The result table from the recorded run, with room for your notes.
- The twenty cases and the blank confusion matrix: fill it by hand. Then the small table at thresholds 0.50, 0.20 and 0.15.
- Your task: the costlier mistake and a first UGX figure for each.

Fill in

My task, in one line	
Which mistake costs more, a false positive or a false negative	
Rough UGX cost of a false positive	
Rough UGX cost of a false negative	
The first question I will put to a vendor	

Think about

Where have I seen a single accuracy figure used to sell something?

What would a majority-class baseline or a simple rule be for the task I chose?

My decision

The costlier mistake for my task, and the first number I will ask a vendor for.	

I keep: The vendor question sheet: ten questions about the numbers behind any AI claim. Keep it next to every proposal you read.

Session 3: worksheet for the paper exercise

A smoke alarm, twenty cases and the confusion matrix

Warm-up (optional): a smoke alarm

In 1,000 events the alarm gave these results (numbers invented for the lesson). Fill the measures in the margin.

	Alarm rings	Alarm silent
Real fire	TP 16	FN 4
No fire	FP 34	TN 946

Accuracy = (TP + TN) / 1,000		**Precision = TP / (TP + FP)**	
Recall = TP / (TP + FN)		**F1 = 2 x precision x recall / (precision + recall)**	
An alarm that never rings: accuracy		**An alarm that never rings: recall**	

The twenty cases

Twenty invoices from the recorded run. Flag an invoice at a score of 0.20 or more. Late means paid late in fact.

No.	Invoice	Amount, UGX	Score	In fact	Flagged? (Y/N)	Cell (TP FP FN TN)
1	INV0318	470,000	0.33	On time		
2	INV0168	744,000	0.47	Late		
3	INV1940	822,000	0.10	On time		
4	INV1486	427,000	0.01	On time		
5	INV0216	463,000	0.07	On time		
6	INV1334	195,000	0.13	On time		
7	INV2427	774,000	0.17	Late		
8	INV1848	824,000	0.06	On time		
9	INV1345	218,000	0.02	On time		
10	INV0761	216,000	0.08	Late		
11	INV0002	524,000	0.08	On time		
12	INV2938	285,000	0.03	On time		
13	INV2286	464,000	0.11	On time		
14	INV1810	309,000	0.13	On time		
15	INV2788	889,000	0.09	Late		
16	INV0512	4,262,000	0.73	Late		
17	INV1030	224,000	0.03	On time		
18	INV1920	268,000	0.12	On time		
19	INV0655	697,000	0.53	Late		
20	INV1629	391,000	0.32	On time		

The confusion matrix and two measures

	Flagged	Not flagged
In fact late	TP (true positive)	FN (false negative)
In fact on time	FP (false positive)	TN (true negative)

The four counts add up to 20. Precision = TP / (TP + FP) = _____. Recall = TP / (TP + FN) = _____.

Move the threshold

Threshold	Flagged	TP	Precision	Recall
0.50				
0.20				
0.15				

Which threshold would you choose, and why?

_____

_____

Session 4

Words as numbers: bag of words, TF-IDF, embeddings and distance

How can a computer find the right clause or policy when I describe it in plain words?

Your steps for today

- The cheat sheet: read each formula and name every symbol (a_1, b_1, Σ, $\|a\|$, $\sqrt{}$, θ, tf, df, N, ln).
- Warm-up: three friends rate two films. Ana (3, 4), Ben (6, 8), Chloe (4, 3). Work out the lengths (5, 10, 5), then the four measures from Ana to Ben and from Ana to Chloe (answers 5 and 1.41, 7 and 2, 50 and 24, 1.00 and 0.96).
- Bag of words: the three tiny messages S1 "send the invoice", S2 "the invoice is late", S3 "the rice is late". Count the six words, then the cosine of S2 with S1 (0.58) and with S3 (0.75).
- The three viewers A (3, 4), B (3, 1), C (6, 0) and the empty table. Fill in the twelve answers by hand.
- The recorded search results for the three kinds of text, with space for your notes.
- Three plain descriptions from your own work and your predictions.

Fill in

Three formal terms from my work	
A plain description of each, using none of its words	
Would a keyword search find it (yes, no, unsure)	
Would a semantic search find it (yes, no, unsure)	
What a wrong first answer would cost me	

Think about

Where do I search a pile of text today, and what do I type?

Where must a search match exactly?

My decision

The pile of text where semantic search would help most, and what a wrong first answer would cost.	

I keep: The cheat sheet: the four measures of closeness (Euclidean distance, Manhattan distance, dot product, cosine similarity), the length of a vector and TF-IDF, each with its formula, every symbol named and the worked numbers.

Session 4: worksheet for the paper exercise

Four measures of closeness

Four measures of closeness between two vectors. Euclidean distance: square root of (gap1 squared + gap2 squared). Manhattan distance: size of gap1 + size of gap2. Dot product: first x first + second x second. Cosine similarity: dot product / (length of the first x length of the second).

Warm-up (optional): three friends rate two films

Ana (3, 4), Ben (6, 8), Chloe (4, 3). Lengths: Ana 5, Ben 10, Chloe 5.

Measure	Ana to Ben	Ana to Chloe
Euclidean distance		
Manhattan distance		
Dot product		
Cosine similarity		

Three viewers

A is 3 across and 4 up. B is 3 across and 1 up. C is 6 across and 0 up. Lengths: A 5, B 3.16, C 6. Mark them on squared paper and draw an arrow from (0, 0) to each.

Measure	A to B	A to C	B to C
Euclidean distance			
Manhattan distance			
Dot product			
Cosine similarity			

Which pair does each measure call closest? Smaller distance, bigger dot product or cosine = closer. _____

Three searches from my own work

Formal term	Plain description (none of its words)	Word search finds it?	Meaning search finds it?

Session 5

Language models: tokens, attention and next-word probability

How does a language model turn words into an answer, and why can the same question give different answers?

Your steps for today

- Write a sentence from your work. Estimate its tokens, then check with the model if it is run live.
- Work the softmax by hand with scores 3, 2 and 0: e to each score, add, divide each by the total.
- Divide the scores by the temperature 2 (giving 1.5, 1 and 0) and repeat. Compare.
- Extra: if the right token was weeks, compute its loss, minus ln of its probability at temperature 1.
- Write one task where variety helps and one where it hurts.

Fill in

My sentence and its token count	
Softmax probabilities at temperature 1 (scores 3, 2, 0)	
Softmax probabilities at temperature 2 (scores 1.5, 1, 0)	
Loss if weeks was right	
A task where variety helps	
A task where it hurts	

Think about

Where in my work would a confident but wrong answer cost the most?

Which of my documents would a model need to see before it could answer questions about my company?

My decision

The task I choose, the category (draft, draft and check, needs a source, keep it human) and why.	

I keep: A one-page sheet: tokens, logits, softmax, temperature, the next-token loss, self-attention and five questions to ask about any answer.

Session 5: worksheet for the paper exercise

A softmax and a temperature by hand

Turn logits into probabilities with softmax. Raise 2.72 to each logit, add the three answers, and divide each answer by the total. The temperature divides the logits first.

Temperature 1 (logits as they are)

Token	Logit	2.72 to the logit	Probability
days	3		
weeks	2		
minutes	0		
Total			

Temperature 2 (divide each logit by 2 first)

Token	Logit / 2	2.72 to the logit	Probability
days	1.5		
weeks	1		
minutes	0		
Total			

Which temperature gives the safer, more repeatable answer, and which gives more variety?

Session 6

Foundation models: pretraining, post-training, model families, reasoning and multimodal models, benchmarks

What is a foundation model, how does it become an assistant, and what does a benchmark score tell me?

Your steps for today

- Average loss: three token probabilities 0.18, 0.50 and 0.25. Write −ln p for each (1.71, 0.69, 1.39), add them (3.79), divide by 3 (1.26).
- Perplexity: a blind guess among 4 words gives loss −ln 0.25 = 1.39 and perplexity e^1.39 = 4. Our 1.26 gives e^1.26 ≈ 3.5.
- Explain supervised fine-tuning and preference tuning (RLHF) in your own words: the data each uses and what changes.
- Accuracy: 712 right of 1,000 questions. Write 712 ÷ 1,000 as a percentage (71.2%) and the four-option guessing baseline (25%).
- Answer the four model card questions for the GPT-2 card, then fill one row of the comparison grid for your own work.

Fill in

Average loss of three tokens	
Perplexity in words	
SFT: data and what changes	
Preference tuning (RLHF): data and what changes	
My task in one line	
Models I can reach	
Three checks	
Ten test cases (list them)	
Model card: licence, size, limits	

Think about

Which claim from a vendor would I now ask to see tested on my own sample, and with which settings?

Where in my work would an invented promise cost the most?

My decision

The route I choose (hosted assistant, reasoning model, open weights or wait for evidence), my task and my three checks.	

I keep: A one-page model comparison grid for your own work, with the questions to ask of a model card and a benchmark claim.

Session 6: worksheet for the paper exercise

Read a model card

Read the printed model card (GPT-2, from its Hugging Face page) and answer four questions with a line from the card.

Question	Line from the card	Found? (Y/N)
What is the licence, and may you use it at work?		
How big is it, and what was it trained on?		
What does the maker say it should not be used for?		
What is missing that you would want to know?		

Session 7

Open weights, file formats, size and running cost

What does it mean to own, host or rent a model?

Your steps for today

- Warm-up: a photo of 12 million pixels, 3 colour numbers each, 1 byte each. How many MB?
- Bits: how many patterns do 4 bits give? And 8 bits (one byte)?
- Multiply the parameters of a 7 billion model by 2 bytes (FP16) and by half a byte (4-bit): memory ≈ N × b.
- Quantise 0.42 with s = 0.1 and z = 8: divide by s, round, add z to get q; read back $\hat{x} = s\,(q - z)$; write the error.
- Break-even: F = 3,000,000 a month, p = 1,000, v = 200. Work out F ÷ (p − v).
- Name one task of yours and its data, choose rent, host or own, and write the licence question you would ask first.

Fill in

Photo sum: 12 million x 3 x 1 byte = ___ MB (answer 36)	
Patterns: 4 bits ___, 8 bits ___ (answers 16 and 256)	
Memory sum at FP16 (7 billion x 2 bytes)	
Memory sum at 4-bit (7 billion x 0.5 byte)	
Quantise 0.42 (s = 0.1, z = 8): x ÷ s = ___, round = ___, q = ___, $\hat{x}$ = ___, error = ___ (answers 4.2, 4, 12, 0.4, 0.02)	
Break-even: 3,000,000 ÷ (1,000 − 200) = ___ requests a month (answer 3,750)	
My task and its data	
My route (rent, host or own) and my reason	
The licence question I would ask first	

Think about

Which of my tasks touch data that should not leave the company?

What volume would make me move from renting to hosting?

My decision

My route for my first model, and the one thing that would change my mind.	

I keep: A one-page rent, host or own decision sheet, with the memory sum, the break-even sum and the licence questions to ask.

Session 7: worksheet for the paper exercise

The memory sum and the decision sheet

Size = how many numbers x bytes for each number. 1 million bytes is 1 megabyte (MB). 1 billion bytes is 1 gigabyte (GB).

Part A: everyday sums

Item	How many numbers	Bytes each	Size in MB
A photo of 12 million pixels, 3 colour numbers each		1	
One minute of music, about 5.3 million numbers	5.3 million	2	
A spreadsheet of 1 million numbers	1 million	8	

Part B: the memory sum for a model

Parameters	At FP16 (x 2 bytes), GB	At 4-bit (x 0.5 byte), GB
1 billion		
7 billion		
70 billion		

Part C: the decision sheet

One task of mine and its data: rent, host or own, my reason, and the licence question I would ask first.

Task and the data it touches	
Route (rent, host or own)	
Reason	
First licence question	

Session 8

Prompting and reasoning models

How do I get reliable work out of an assistant?

Your steps for today

- Take a prompt you use often and write it with five parts: role, context, examples, format and constraints.
- Write three cases with known answers to form an evaluation set.
- Run it, score it against the known answers (accuracy = correct ÷ total) and fill one row of the prompt log.
- Change one thing and run it again.

Fill in

My prompt, rewritten with role, context, examples, format and constraints	
Three evaluation cases with known answers	
Version, what changed, runs, how many right, accuracy = correct ÷ total	
Who checks the figures before I use them	

Think about

Which of my prompts has a format I could ask for as a table or JSON?

Where could a calm, confident answer hide a slip of the size we saw today?

My decision

The prompt I will test first, and who checks it.	

I keep: A reusable prompt template and a prompt log, so that every prompt you keep has been tested on an evaluation set.

Session 8: worksheet for the paper exercise

The five-part prompt

Rewrite a weak prompt as five parts: role, context, examples, format and constraints (and what to say when unsure).

Part A: 'Make dinner.'

Part	My sentence
Role	
Context	
Examples	
Format	
Constraints	

Part B: 'Reply to this customer complaint.'

Customer: Mercy Atim, whose jerrycans of oil leaked for the third time (a Nile Valley Foods example).

Part	My sentence
Role	
Context	
Examples	
Format	
Constraints	

Part C: label three mobile money SMS

SMS (the first two are the examples)	Label
'You have received UGX 20,000 from Mary.'	Money in
'You have sent UGX 50,000 to Peter.'	Money out
'You have received UGX 8,000 from John.'	

Part D: prompt log

Version	What changed	Runs	How many right	Next change

Part E: my evaluation set of three cases

Three cases for which I know the right answer, then the result of one run.

Case	Input	Known answer	Result of run 1	Right? Y/N
Case 1				
Case 2				
Case 3				

Who checks the figures before I use the answer? _____

Session 9

Giving a model your knowledge: retrieval-augmented generation, long context and fine-tuning

How do I give an AI model my company's knowledge, and when is each way worth it?

Your steps for today

- For each of the six NVF requests, write P (prompting), R (RAG) or F (fine-tuning) and one line of reason.
- Write ten questions with known answers for one document in your job.
- Run the cheapest rung on your ten questions and count the right answers. Optional.

Fill in

The request from my job that needs my knowledge	
The method I think it needs (prompting, RAG, fine-tuning)	
My ten questions and where each answer is found	
Right answers on the cheapest rung (out of 10)	
What would make me move to the next method	

Think about

Which of my documents change often, and who would keep an index up to date?

Do I hold examples I may use for fine-tuning, with personal data removed?

My decision

The method I will try first, and the ten-question test I will write.	

I keep: A one-page decision tree: prompting first, then retrieval-augmented generation (RAG), then fine-tuning, and the test that tells you when to move on.

Session 9: worksheet for the paper exercise

Choose the method

Choose the method for each request: P (prompting), R (RAG, retrieval-augmented generation) or F (fine-tuning). Give one line of reason and the first test you would run.

	Request	Method (P, R or F)	Reason (one line)	First test
a	Summarise one supplier email.			
b	Answer staff questions from a handbook that changes each quarter.			
c	Every customer reply ends with the house sign-off.			
d	Check a price against one pasted quote.			
e	Answer questions about 400 contracts.			
f	Draft replies in the Sales Manager voice from 200 past emails.			

The ten-question test I would run first on the cheapest method: write it on the Session 10 evaluation set page or on paper.

Session 10

When models fail, and how to check them

How do I find out when an AI answer is wrong, before it costs me?

Your steps for today

- Label the practice answer sheet against the key: right, wrong, missed, invented; accuracy = correct ÷ 10.
- Write your own ten-question evaluation set for one document in your job (question, ground truth, source, cost of a wrong answer).
- Run it on one assistant, twice, and score it. Optional.

Fill in

The document I chose	
My ten questions, ground truth and sources	
Score on run 1 and run 2 (right, wrong, missed, invented)	
The cost of one wrong answer, in UGX or in time	
Expected cost = P(error) × cost of one error	
Who signs off the answers	

Think about

Which of the five cases could happen in my own work this month?

Which question in my set would catch an invented answer?

My decision

The level of checking for one AI use in my job, and the first check I will make.	

I keep: A five-step check routine and an evaluation-set template (questions with ground truth), each on one page.

Session 10: worksheet for the paper exercise

Score an answer sheet and write an evaluation set

Mark each answer on the practice sheet against the ground truth: right, wrong, missed or invented. The key is handed out in class.

Q	Question	Sheet says	Ground truth	Mark
Q01	How many working days of paid annual leave does a full-time staff member earn in each leave year?	21 working days (clause 2.1).		
Q02	How long before the first day of leave must a leave request reach the line manager?	At least 14 days before the first day of leave (clause 2.2).		
Q03	Who must approve an expense claim of UGX 1,500,000?	The Managing Director (clause 3.2).		
Q04	Within how many days of spending must receipts reach the Finance office?	Within 14 days of the spending (clause 3.1).		
Q05	How is approved overtime paid on a Sunday?	1.5 times the normal hourly rate (clause 1.3).		
Q06	May a staff member paste a customer price list into a public AI tool?	No, unless the Managing Director has approved the tool in writing (clause 4.4).		
Q07	How soon must a lost company laptop be reported, and to whom?	Within 4 hours of discovery, to the Operations Manager (clause 5.3).		
Q08	What is the most the company will pay for one night of accommodation on company travel?	UGX 180,000 a night (clause 3.3).		
Q09	How many days of paid sick leave does a staff member get each year?	10 days (clause 2.4).		
Q10	What rate per kilometre does the company pay when a staff member uses a private car for work?	Not stated in the handbook.		
	Count	Right ___ Wrong ___ Missed ___ Invented ___		

Which error would cost most, and why?

_____

My ten-question evaluation set

Id	Question	Ground truth	Source	Run 1	Run 2	Cost of a wrong answer
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
Q7						
Q8						
Q9						
Q10						

Two of the ten questions should have no answer in the document. The right answer to those is "not stated".

Session 11

Agentic AI: tools, loops, MCP, autonomy and guardrails

What changes when an assistant can act as well as answer?

Your steps for today

- Fill the four rows of a Tool Boundary Sheet (the permissions table) for the mailbox assistant.
- Name the row that stops M02, M03, M07 and M08.
- Trace plan, act and observe for M01, M03 and M08, as three audit log lines.
- Count the approvals and your minutes: approval time T = n × t (n approvals, t minutes each).
- For your own task, write likelihood × impact for the riskiest permission.

Fill in

The task I would give an agent	
The autonomy level I would allow	
Permissions it needs (read, write, send, spend)	
Who approves (human-in-the-loop), and the off switch	
Approval minutes a week, T = n × t (approvals × minutes each)	

Think about

Which app that I already use asks for permissions it does not need?

Who would notice first if an agent in my company went wrong?

My decision

The autonomy level I allow for my task, and the one permission I will never give.	

I keep: Your Tool Boundary Sheet, the permissions table for an agent: for read, write, send and spend, what it may do, what a person approves (human-in-the-loop) and what it must never do, with the owner, the audit log and the off switch.

Session 11: worksheet for the paper exercise

The permissions table and the agent loop

Fill the four rows of the permissions table for the mailbox assistant (all mails in the demonstration mailbox are invented).

Row	It may	A person approves	It must never
Read			
Write			
Send			
Spend			

Mail	What it asks	Row that stops the assistant
M02	Invoice with no purchase order	
M03	Credit of UGX 180,000	
M07	Price rise of 4 percent	
M08	An instruction hidden in a statement	

The agent loop for three mails

Mail	Plan	Act (one tool)	Observe, and where the loop must stop
M01			
M03			
M08			

Approvals a week x minutes each = _____ minutes a week. Owner: _____ Off switch: _____

Session 12

Workflow mapping (practical)

Which of my tasks should be a rule, an AI step or an agent?

Your steps for today

- Map one weekly task in lanes: trigger, steps, handoffs, check.
- Tag each step: rule-based automation, AI step, agent or person, with a reason and a check.
- Sketch one assistant with its four permission rows, if one fits.
- Fill the Inputs sheet of the Business Case Workbook (your baseline and the new routine) and mark E or M.

Fill in

My workflow and its trigger	
The handoff that causes the most waiting	
The step I would hand to AI first, and how I check it	
My baseline: minutes each time today, times a month	
Hours saved a month (runs × minutes saved ÷ 60), net benefit a month (benefits − costs) and payback period (set-up ÷ net a month)	
The four numbers I will record for four weeks	

Think about

Which step on my map is a rule I could set up this week without AI?

Who else in my organisation runs a workflow like mine?

My decision

What I build or test first, and the number that will show it worked.	

I keep: Your mapped workflow, the choice for each step, your assistant sketch, and the baseline and hours-saved estimate in the Business Case Workbook.

Session 13

Governance and buying (practical)

What rules must my company set for AI, and how do I question what I am sold?

Your steps for today

- Edit the Risk Register for your own work: a score, an owner and a control for your top five.
- Draft sections 3, 4, 5, 7, 8 and 10 of the AI Use Policy Template.
- Answer three groups of the Vendor Question List for one tool, and write five questions with red-flag answers.
- Fill in the lawyer page and name the policy owner, a deputy and the review date.

Fill in

My work and the tool I chose	
My top five risks, their scores and levels	
Owner and control for each	
Approved tools in my policy, each with an owner	
What may go in at each data classification level	
What always needs human oversight by a named reviewer	
Who hears about an incident, and how fast	
My three biggest gaps about the tool	
My five vendor questions and the red-flag answer for each	
Which of my AI uses involve personal data	
Policy owner, deputy and review date	

Think about

Which risk surprised me when I scored it?

Who in my company already uses AI through a personal account?

Which vendor claim would I have believed before today?

My decision

Decline, ask in writing or run a pilot: my route for the next proposal, the reason and the first step.	

I keep: Your Risk Register, your AI Use Policy draft, your Vendor Question List with five questions, and your page of questions for the lawyer.

Session 14

Data readiness assessment, roadmap and 90-day plan (practical)

How ready is my organisation's data for AI, what will I fix first, and who will own it?

Your steps for today

- Build the data inventory in Information Inventory.xlsx: at least six sources, each owned and classified.
- Score the six areas Green, Amber or Red with evidence, and read the total out of 12.
- Turn your top three gaps into at least six roadmap actions with owners, dates, hours and KPIs.
- Put the day 30, 60 and 90 check-ins in your diary and list three risks to the plan.

Fill in

My organisation and my plan start date	
Sources in my inventory (number) and how many are Confidential or Personal	
My six ratings and my total out of 12	
My top three gaps, with evidence and a first fix	
My roadmap actions, each with owner, due date, hours and KPI	
Decisions the company must make, who decides and by when	
Three risks to the plan and what I will do	
My check-in dates: day 30, day 60, day 90	
How many gaps I will close by day 90	
The action I will take within 24 hours	

Think about

Which area surprised me when I scored it?

Which of the fourteen sessions changed how I think most, and why?

What do I now understand about AI before I decide anything about it?

Who must hear about this plan first, and what do I want from them?

My decision

How many gaps my roadmap closes by day 90, and my first action within 24 hours.	

I keep: Your inventory and scorecard, your roadmap with its risks and decisions, the Follow-up Kit (Check-in Form, Results Tracker, Sponsor Briefing) and your answer to the Session 0 question.

Words used in this course

Word	What it means	First used
Model	A function with adjustable parameters that has learned patterns from data and makes predictions.	Session 0
Parameters	All the weights and biases of a model: the numbers that training adjusts.	Session 0
Training	Repeating predict, measure the loss, update the parameters, until the loss is small.	Session 0
Inference	Using a trained model to make a prediction or generate an answer.	Session 0
Overfitting	When a model fits its training data, noise included, and does worse on new data.	Session 0
Pixel	One square of a picture, stored as a number (grey) or three numbers (colour).	Session 1
Feature	An input measurement, such as the floor area of a house or the ink in the middle of a digit.	Session 1
Neuron	A unit that computes a weighted sum of its inputs, adds a bias and applies an activation function.	Session 1
Weight	A number that says how much one input counts. Training adjusts the weights.	Session 1
Layer	A set of neurons that take the same inputs. Layers between input and output are hidden layers.	Session 1
Decision tree	A model made of yes-or-no questions on features, ending in an answer called a leaf.	Session 2
Gini impurity	A number for how mixed the answers in a group are: 0 when every answer is the same.	Session 2
Cross-validation	Cut the data into k folds; each fold takes a turn as the held-out test; average the k scores.	Session 2
k-nearest neighbours	Predict from the k most similar past examples, found by measuring a distance.	Session 2
Confusion matrix	A table that counts the four kinds of outcome of a classifier: true positives, false positives, false negatives and true negatives.	Session 3
Precision and recall	Precision: of the cases flagged, the share that were real. Recall: of the real cases, the share that were flagged.	Session 3
Threshold	The score at or above which the model says positive. You choose it, and moving it trades false alarms for misses.	Session 3
ROC curve and AUC	ROC: the true positive rate against the false positive rate, one point per threshold. AUC: the area under it, from 0.5 (coin flip) to 1.	Session 3
Baseline	A simple reference model to beat, such as the majority-class baseline that always predicts the most common class.	Session 3
Vector	A list of numbers in a fixed order, such as Ana's two film ratings (3, 4). Each position is a dimension. Drawn as an arrow from 0.	Session 4
Cosine similarity	A score from −1 to 1 for how closely two vectors point the same way: the dot product divided by both lengths.	Session 4
Bag of words	A message as a vector of word counts, one dimension for each word in the vocabulary. The word order is lost.	Session 4
TF-IDF	Term frequency times inverse document frequency: a word count weighted so that rare words count more than common ones.	Session 4
Embedding	A vector, produced by a trained model, whose numbers place texts with similar meanings close together.	Session 4

Word	What it means	First used
Token, token id	A token is a piece of text the model reads. Its token id is its number in the fixed vocabulary.	Session 5
Logit	The raw score the model gives each possible next token. Higher means more likely.	Session 5
Softmax	Turns the scores into probabilities: e to each score, divided by the total.	Session 5
Temperature, sampling	Sampling draws the next token by chance. Temperature divides the scores first: low sharpens, high flattens.	Session 5
Self-attention	Each token weighs the earlier tokens, using a query, a key and a value.	Session 5
Pretraining, base model	Next-token prediction on a text corpus, a huge collection of text. The result, a base model, continues text.	Session 6
Perplexity	e raised to the average next-token loss: as unsure as choosing among that many equally likely tokens. Lower is better.	Session 6
Post-training: SFT, RLHF	Supervised fine-tuning on answers people wrote, then preference tuning, such as RLHF, on answers people compared.	Session 6
Model families	Designs built on the same idea: mixture of experts, reasoning, multimodal, diffusion and small language models.	Session 6
Benchmark, accuracy	A fixed set of questions with an answer key and a scoring rule. Accuracy = correct ÷ total.	Session 6
Bit, byte, precision	A bit is a 0 or a 1; 8 bits make a byte. Precision is how many bits each parameter is stored in: FP32, FP16, INT8, 4-bit.	Session 7
Quantisation	Storing each parameter with fewer bits, by rounding it to one of fewer levels, so the file shrinks.	Session 7
File format	The container that packs the parameters into a file: Safetensors, GGUF or ONNX. It does not set the precision.	Session 7
Open weights	The trained parameters can be downloaded, under a licence that can still limit use. Open source (OSI) needs more.	Session 7
Prompt, system prompt	The prompt is all the text you give a model. A system prompt holds standing instructions; a user prompt is this request.	Session 8
Zero-shot, few-shot	Zero-shot prompting gives only the instruction. Few-shot prompting adds a few worked examples (shots) first.	Session 8
Structured output, JSON	An answer in fixed named fields. JSON is a plain-text format of named fields that a program can read.	Session 8
Evaluation set, accuracy	An evaluation set is test cases with known right answers. Accuracy = correct answers ÷ total cases.	Session 8
Chain-of-thought, reasoning model	Chain-of-thought asks the model to write its steps. A reasoning model writes its own working before it answers.	Session 8
Retrieval-augmented generation (RAG)	A retriever fetches relevant text from your documents, and the model answers from it.	Session 9
Chunk, embedding, cosine similarity	A chunk is a piece of a document. An embedding is a vector for its meaning. Cosine compares two vectors.	Session 9
Top-k retrieval, vector database	Keep the k chunks with the highest scores. A vector database stores the embeddings and finds the closest fast.	Session 9
Context window	The most tokens a model can read and write at one time. Long context means a large window.	Session 9

Word	What it means	First used
Fine-tuning, LoRA, rank	Fine-tuning is more training on your examples. LoRA (low-rank adaptation) trains a small update; rank sets its size.	Session 9
Hallucination	A fluent, confident output that the facts and sources do not support, such as an invented citation.	Session 10
Prompt injection	Text inside the input written to steer the model. It is data, never an order.	Session 10
Bias	Results that favour or miss some groups, often learned from skewed past examples.	Session 10
Evaluation set, ground truth	Questions kept for scoring, each with its known correct answer (the ground truth).	Session 10
Expected cost of an error	What errors cost on average: the probability of an error times the cost of one error.	Session 10
Agent	A model that answers, given a goal and tools, which chooses its own next step in a loop of plan, act and observe.	Session 11
Tool use (function calling)	The model writes a request to run a named tool, such as read_mail; a program runs it and returns the result.	Session 11
Least privilege	Give an agent the smallest set of permissions (scopes) that still lets it do the job.	Session 11
Human-in-the-loop	A person approves, edits or rejects an action before it runs.	Session 11
Workflow map	A drawing of a repeated task: one lane per person, a box per step, arrows for order and handoffs, from trigger to check.	Session 12
Rule-based automation	A machine runs fixed rules with no judgement, such as a bank standing order.	Session 12
AI step	One call to a model that makes a judgement, such as spam or not spam, inside a fixed path.	Session 12
Payback period	How long until the savings repay the one-off cost: one-off cost divided by net saving a month.	Session 12
Return on investment (ROI)	Net benefit (benefits minus costs) divided by costs, times 100 percent, over a stated period.	Session 12
AI governance	The policies, roles and checks by which an organisation directs and controls its use of AI.	Session 13
Acceptable use policy	A written rule on who may use which AI tools, for what, and with what data.	Session 13
Risk register	A table of what could go wrong, scored by likelihood and impact, with an owner and a control.	Session 13
Data classification	Sorting information into levels, such as public or confidential, each with its own handling rule.	Session 13
Human oversight	A named person who checks, approves or can stop what an AI system does.	Session 13
Vendor due diligence	Checking a supplier's claims, security and contract terms before you commit.	Session 13
Data inventory	A written list of the data sources you hold, with owner, location and who may see each.	Session 14
Data quality dimensions	The standard checks: completeness, accuracy, consistency, timeliness, uniqueness and validity.	Session 14
Metadata and lineage	Metadata is data about the data, such as date, owner and version. Lineage is the recorded path of data from its source to its use.	Session 14

Word	What it means	First used
Data governance	The policies, roles and rules that decide who may do what with which data.	Session 14
Roadmap and KPIs	A plan of actions with owners and milestones. A KPI, key performance indicator, is a number that shows progress.	Session 14
Evaluation set	Questions with known answers, used to measure how well a system performs.	Session 14